

# Assignment 2-Task2

## Implementing Linear Regression Using a Multi-Layer Neural Network From Scratch

### Objective

Predict **MedHouseVal** using two selected inputs, **MedInc** and **HouseAge**, with a fully connected neural network implemented from scratch. The report focuses on architecture, experiments, plots, and measured test performance.

### Dataset & Preprocessing

| Item           | Setting                                                   |
|----------------|-----------------------------------------------------------|
| Dataset        | California Housing — 20,640 samples                       |
| Inputs         | MedInc, HouseAge                                          |
| Target         | MedHouseVal                                               |
| Train / Test   | 16,512 / 4,128 (80 / 20)                                  |
| Normalization  | Training-set z-score statistics applied to train and test |
| Epochs         | 1,000                                                     |
| Loss           | Mean Squared Error (MSE)                                  |
| Initialization | He initialization, seed = 42                              |

### Network Architecture

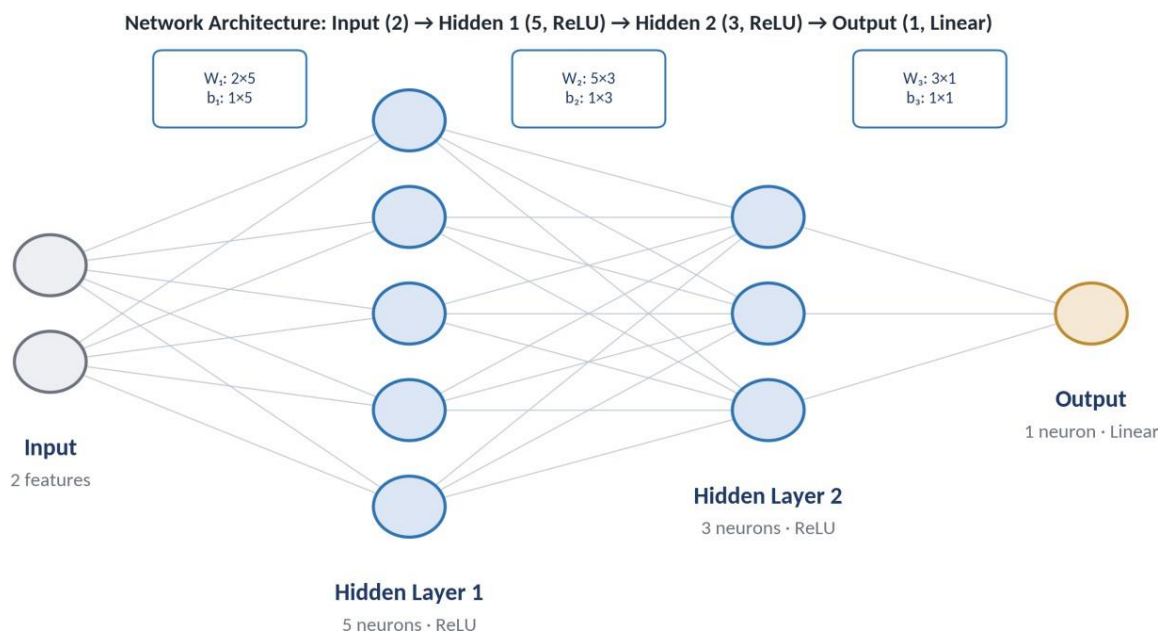


Figure 1. Required architecture: Input (2) → Hidden 1 (5, ReLU) → Hidden 2 (3, ReLU) → Output (1, Linear).

| Layer    | Neurons | Activation | Weights      |
|----------|---------|------------|--------------|
| Input    | 2       | —          | —            |
| Hidden 1 | 5       | ReLU       | $2 \times 5$ |
| Hidden 2 | 3       | ReLU       | $5 \times 3$ |
| Output   | 1       | Linear     | $3 \times 1$ |

## 1. Learning Rate Experiment

Basic Gradient Descent was trained for 1,000 epochs using learning rates 0.01 and 0.001. All other settings were kept unchanged.

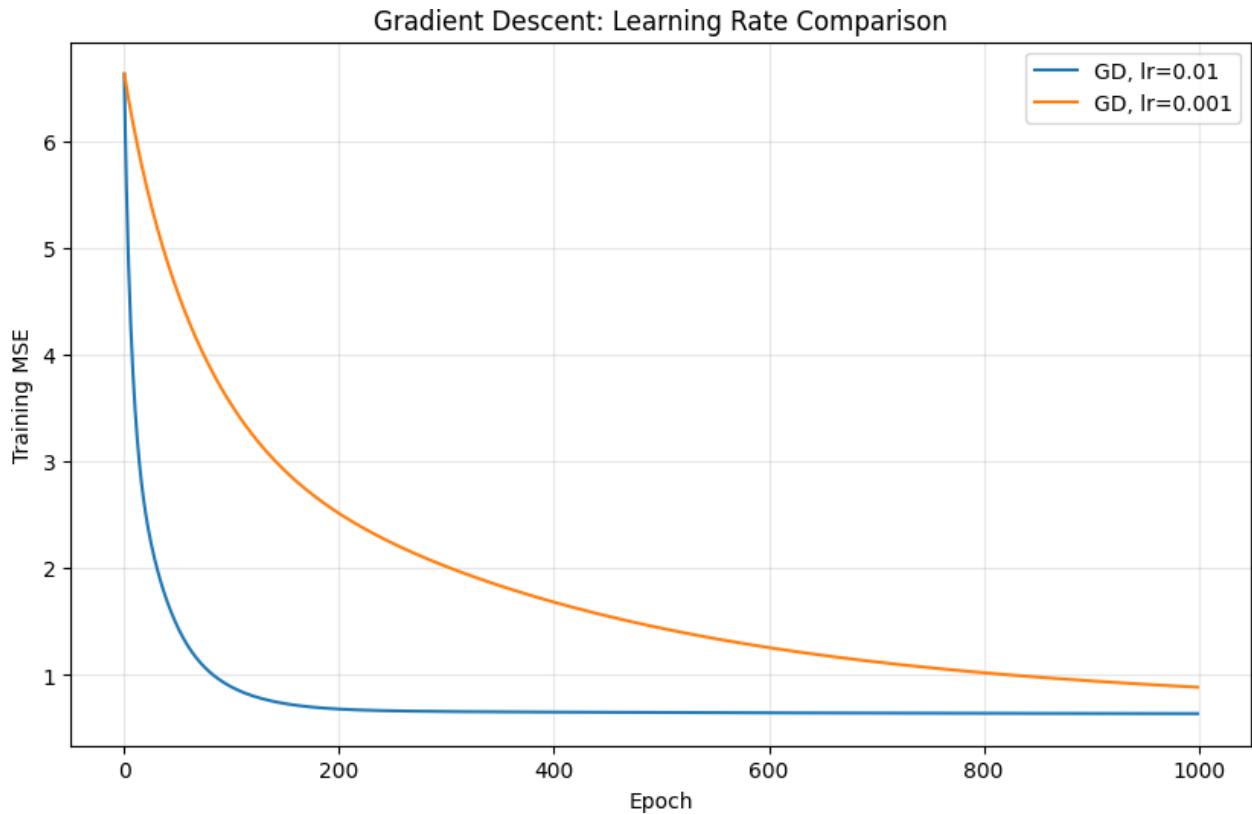


Figure 2. Training MSE for Gradient Descent at learning rates 0.01 and 0.001.

| Learning Rate | Final Train Loss | Test MSE |
|---------------|------------------|----------|
| 0.01          | 0.640194         | 0.648101 |
| 0.001         | 0.887155         | 0.897964 |

**Observation.** Learning rate 0.01 converged faster and reached a lower final training loss and test MSE within the fixed 1,000-epoch budget. The 0.001 run was still descending at the end of training.

**Best learning rate from this experiment:** 0.01.

## 2. Optimizer Comparison

Gradient Descent, Momentum, and Adam were compared from the same initial weights. The first optimizer comparison used learning rate 0.001.

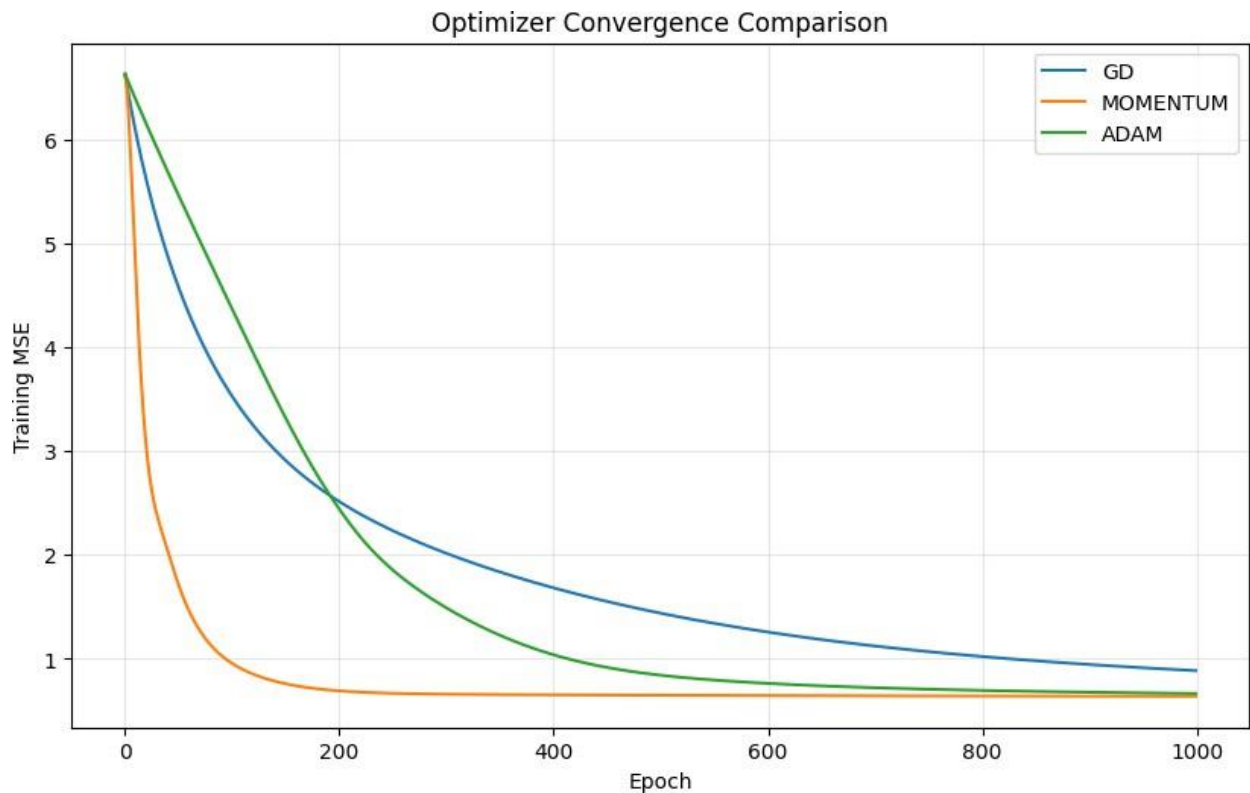


Figure 3. Optimizer convergence at learning rate 0.001.

| Optimizer        | Learning Rate | Final Train Loss | Test MSE |
|------------------|---------------|------------------|----------|
| Momentum         | 0.001         | 0.639867         | 0.647937 |
| Adam             | 0.001         | 0.666758         | 0.683425 |
| Gradient Descent | 0.001         | 0.887155         | 0.897964 |

**Observation.** Momentum gave the lowest test MSE at learning rate 0.001 and converged much faster than plain Gradient Descent. Adam also substantially outperformed plain Gradient Descent.

**Key result:** optimizer choice had a clear effect on convergence and final performance.

### 3. Complete Optimizer × Learning-Rate Results

All six combinations were tested to identify the best overall configuration.

| Rank | Configuration                | Final Train Loss | Test MSE |
|------|------------------------------|------------------|----------|
| 1    | Momentum, lr = 0.01          | 0.610460         | 0.619313 |
| 2    | Adam, lr = 0.01              | 0.614773         | 0.625793 |
| 3    | Momentum, lr = 0.001         | 0.639867         | 0.647937 |
| 4    | Gradient Descent, lr = 0.01  | 0.640194         | 0.648101 |
| 5    | Adam, lr = 0.001             | 0.666758         | 0.683425 |
| 6    | Gradient Descent, lr = 0.001 | 0.887155         | 0.897964 |

**Best overall configuration:** Momentum with learning rate 0.01, using the original 2–5–3–1 network. Test MSE = **0.619313**.

This configuration was selected for the final predicted-versus-actual evaluation.

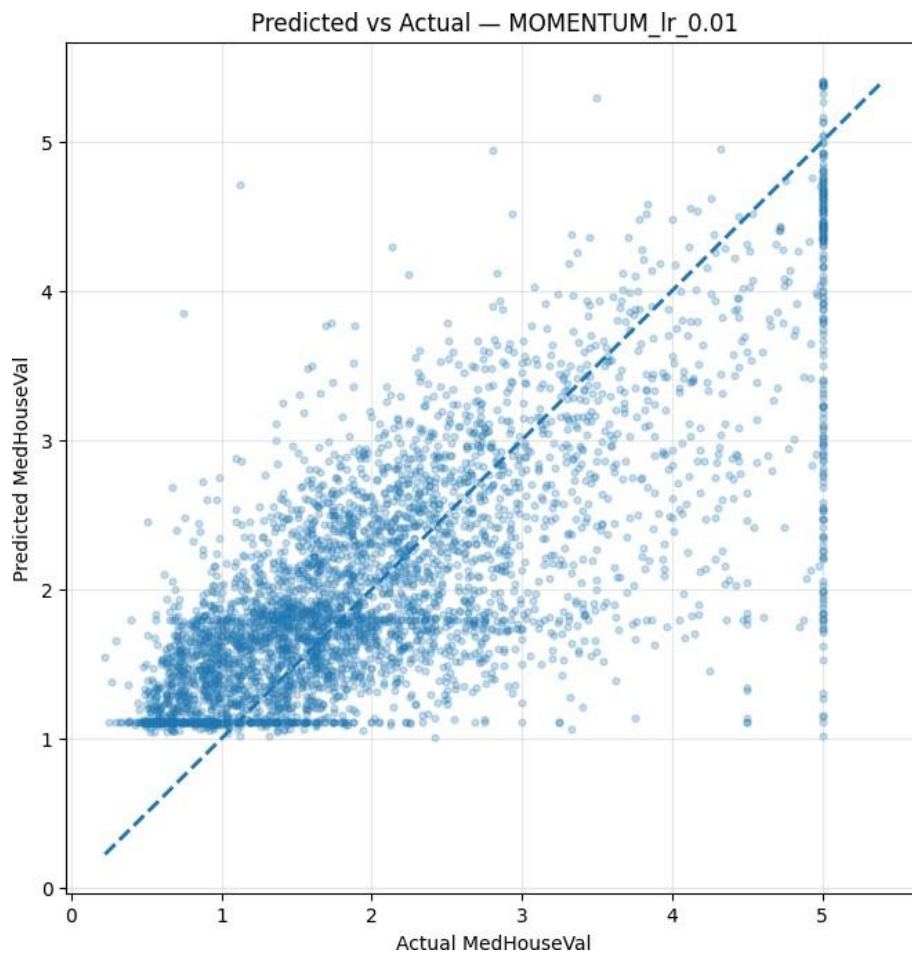


Figure 4. Predicted versus actual MedHouseVal for the best configuration. The dashed diagonal represents perfect prediction.

## 4. Final Prediction Performance

| Metric        | Best Model Value             |
|---------------|------------------------------|
| Test MSE      | 0.619313                     |
| Test RMSE     | 0.786964                     |
| Configuration | 2-5-3-1, Momentum, lr = 0.01 |

**Observation.** The predictions broadly follow the diagonal, showing that the network captures the general relationship between the two inputs and house value. The spread also shows the limitation of using only two features.

## 5. Bonus 1 — Additional Hidden Layer

The original 2-5-3-1 network was compared with a deeper 2-5-3-2-1 network using Adam, learning rate 0.001, 1,000 epochs, and the same initialization seed.

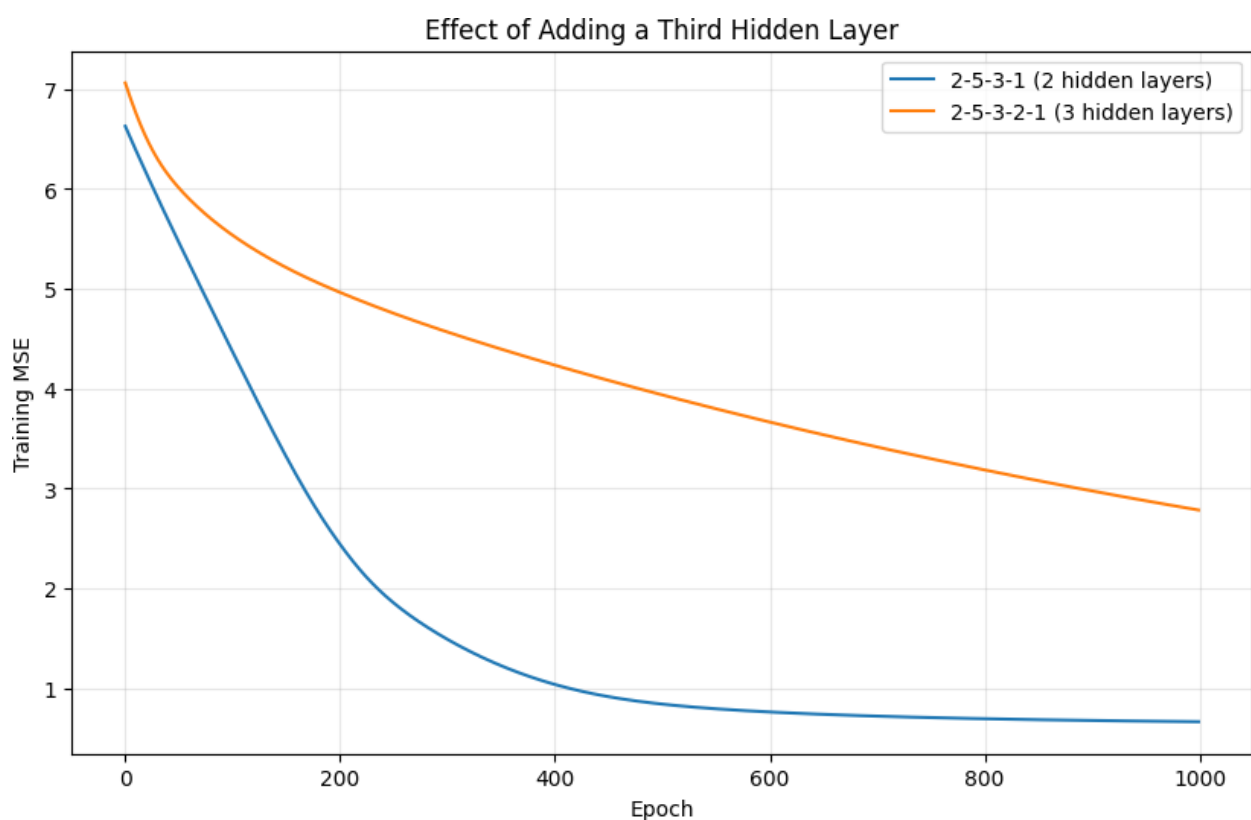


Figure 5. Training loss after adding a third hidden layer.

| Architecture | Hidden Layers | Test MSE |
|--------------|---------------|----------|
| 2-5-3-1      | 2             | 0.683425 |
| 2-5-3-2-1    | 3             | 2.761936 |

**Observation.** The extra hidden layer hurt performance under the same 1,000-epoch budget. The deeper network converged much more slowly and produced more than four times the test MSE.

## 6. Bonus 2 — L2 Regularization

L2 regularization was tested on the original 2–5–3–1 network using Adam, learning rate 0.001, and  $\lambda = 0.001$ . The result was compared with the same model without L2.

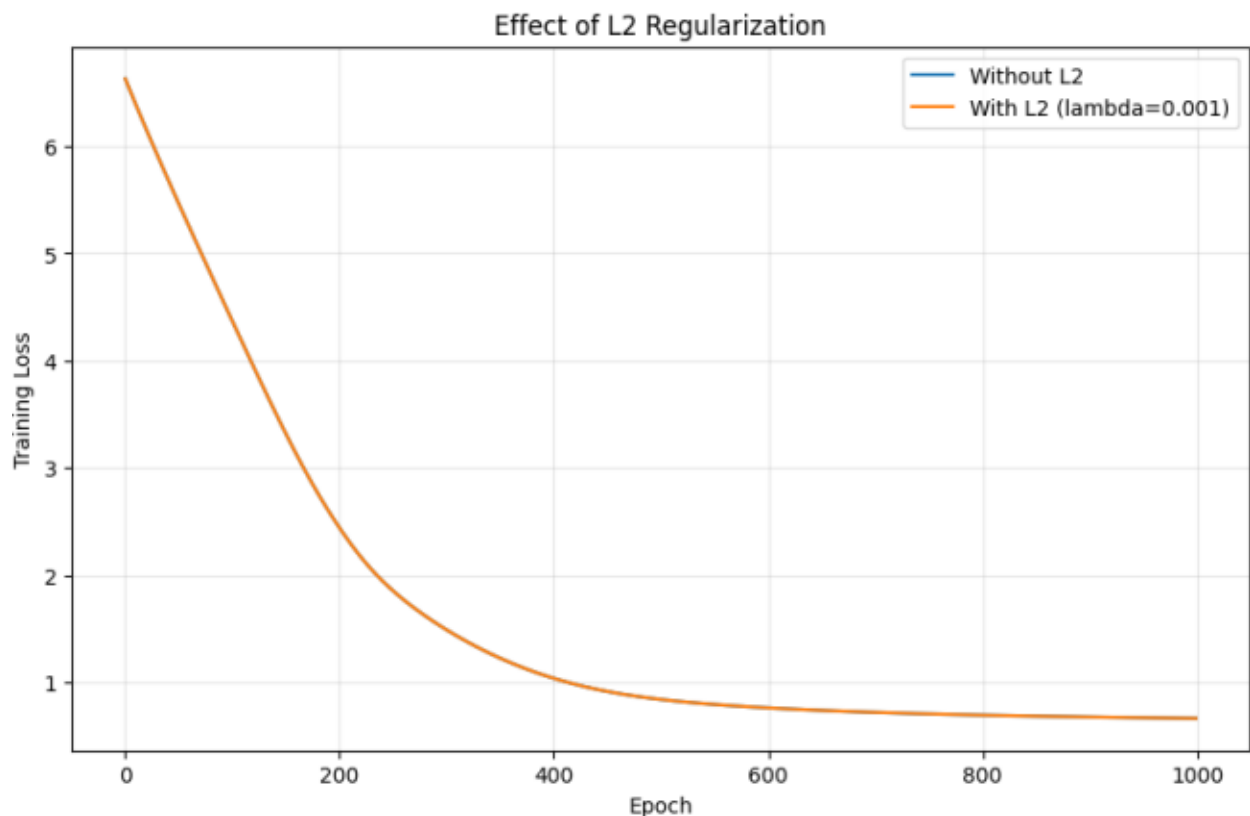


Figure 6. Training loss with and without L2 regularization ( $\lambda = 0.001$ ).

| Configuration              | Final Train Loss | Test MSE |
|----------------------------|------------------|----------|
| Without L2                 | 0.666758         | 0.683425 |
| With L2, $\lambda = 0.001$ | 0.666755         | 0.683425 |

**Observation.** L2 regularization at  $\lambda = 0.001$  had a negligible effect. The loss curves are almost identical and the test MSE is unchanged to six decimal places.

**Bonus conclusion:** Neither additional depth nor the small L2 penalty improved test performance in these experiments.

## 7. Overall Results & Conclusion

The following table summarizes every tested configuration.

| Experiment    | Optimizer | LR    | L2    | Test MSE |
|---------------|-----------|-------|-------|----------|
| Main grid     | Momentum  | 0.01  | 0     | 0.619313 |
| Main grid     | Adam      | 0.01  | 0     | 0.625793 |
| Main grid     | Momentum  | 0.001 | 0     | 0.647937 |
| Main grid     | GD        | 0.01  | 0     | 0.648101 |
| Main grid     | Adam      | 0.001 | 0     | 0.683425 |
| Main grid     | GD        | 0.001 | 0     | 0.897964 |
| Bonus: deeper | Adam      | 0.001 | 0     | 2.761936 |
| Bonus: L2     | Adam      | 0.001 | 0.001 | 0.683425 |

### Final Findings

1. **Best model:** the original 2–5–3–1 architecture with Momentum and learning rate 0.01 achieved the lowest test MSE of **0.619313** (RMSE  $\approx$  0.787).
2. **Learning rate:** 0.01 consistently outperformed 0.001 for the tested optimizers within the 1,000-epoch budget.
3. **Optimizer:** Momentum performed best overall, while Adam at  $\text{lr} = 0.01$  was a close second.
4. **Additional depth:** adding a third hidden layer reduced performance substantially in this experiment.
5. **L2 regularization:**  $\lambda = 0.001$  produced essentially no change in test MSE.
6. **Model limitation:** only MedInc and HouseAge were used, so the network cannot exploit the other available housing features.

### Overall conclusion

For this two-feature regression problem, tuning the optimizer and learning rate produced more useful gains than increasing network depth or applying a very small L2 penalty. The final selected model is compact, fully implemented from scratch, and achieved a test MSE of 0.619313 under the specified experimental setup.